In this chapter, we review the mechanics of searching unstructured data by employing full-text search APIs. Elasticsearch provides a handful of full-text queries in the form of `match`, `query_string`, and others. As `match` queries are the most commonly used queries when working with full text, we dedicate a good chunk of the chapter to various `match` queries. We also work through query-string searches, which are equivalent to using an URI request search but with a request body similar to Query DSL.

10.1 Overview

When we search on a retail website such as Amazon or eBay, we see results close to what we are looking for. If the results are not what we expect, we express our frustration and vow never to return to that site or application, don't we? A user's search experience is crucial to retaining a happy customer.

Relevance is all about how appropriate search results are and how closely they are related to what the user is searching for. Elasticsearch thrives on producing relevant results with speed and accuracy by employing sophisticated relevance algorithms.

When we talk about relevance, two measures usually spring to mind: precision and recall. Relevance is measured using these two factors, and understanding them at a high level is important (although not crucial).

10.1.1 Precision

Precision is the percentage of relevant documents of the overall number of documents returned. When a query returns results, not all results are directly related to the query. There may be a few nonrelevant documents in the results.

For example, suppose we are searching for a 4K television from a particular manufacturer (say, LG), and we obtain 10 results. Not all of these results are relevant: a few are 4K cameras (see figure 10.1), and a couple are projectors, because LG also produces those.

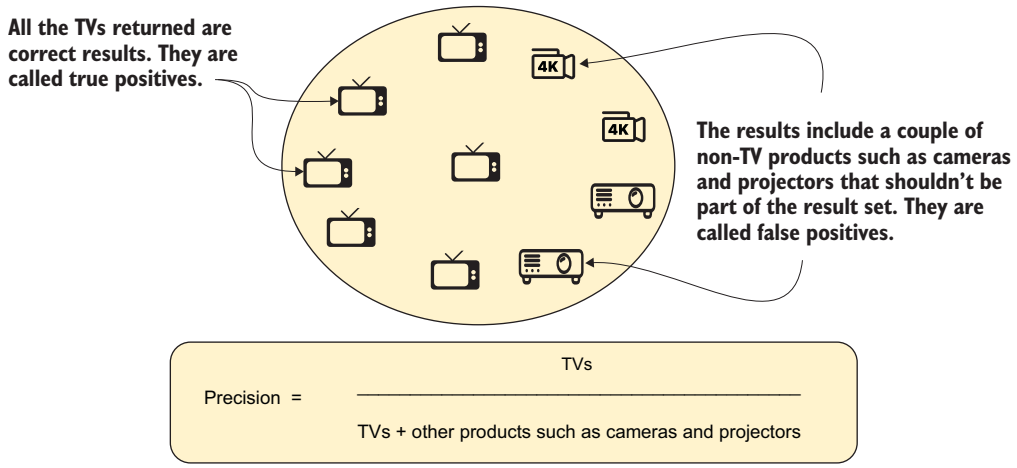


Figure 10.1 A precision example in action, returning search results for a 4K television